Lecture 18

The complete solution set

Uses.

Summary 12: Componentwise operations define real vectors, linear combinations and independence. Standard coordinate entries are unique.
Summary 13: For real columns, two independent directions describe a plane through the origin.
Summary 15: Complex scalars obey the arithmetic laws used in component computations.
Summary 17: Matrix–column action records a linear system. Row operations preserve solutions and expose all free variables. Route-independent pivot counts use the cited uniqueness of reduced form.

Objectives. *One solution of a consistent system determines the location of its whole solution set. The system with zero right sides determines the allowed differences between solutions. We construct a basis for those differences from the free variables and use it to describe every solution without repetition. The dimension convention is stated in terms of the cited intrinsic pivot count.*

By the end of this lecture you will be able to:

- Prove that a consistent system's solutions are one particular solution plus its null space.
- Test independence and span membership by solving coefficient systems.
- Prove that the special free-variable solutions form a null-space basis, including the zero-space case.
- Compute a complete solution family and state the cited input to the rank and dimension convention.

18.1 Homogeneous systems and linear subspaces

Keep $K = \mathbb{R}$ or $K = \mathbb{C}$, and let A have m rows and n columns, with $m, n \geq 1$. Column addition and scalar multiplication in K^n are defined entrywise, just as for the real vectors of Lecture 12. Write 0_k for the column of k zero entries. A system $Ax = 0_m$ is *homogeneous*. Its *null space* (also called its kernel) is

$$\ker A = \{x \in K^n : Ax = 0_m\}. \tag{18.1}$$

The zero column is always a homogeneous solution.

The coefficient action is linear: for $u, v \in K^n$ and $a, b \in K$,

$$A(au + bv) = aAu + bAv. \tag{18.2}$$

For each row index $i = 1, \dots, m$, its left entry is

$$\sum_{j=1}^{n} a_{ij}(au_j + bv_j) = a\sum_{j=1}^{n} a_{ij}u_j + b\sum_{j=1}^{n} a_{ij}v_j,$$

which is the right entry. Equality of all entries proves the formula.

A subset $V \subseteq K^n$ is a *linear subspace* if it contains 0_n and is closed under addition and scalar multiplication. The null space is such a subspace. Indeed, if $Au = Av = 0_m$, linearity gives $A(u + v) = 0_m$ and $A(cu) = 0_m$ for every $c \in K$. Thus the sum and each scalar multiple remain in $\ker A$. This statement concerns the homogeneous system; arbitrary nonzero right sides need a translation.

18.2 Particular plus homogeneous

A *particular solution* is one fixed column p satisfying $Ap = b$. Such a column is available only if the system is consistent. For a linear subspace $V \subseteq K^n$, its translate by $p \in K^n$ is

$$p + V = \{p + h : h \in V\}.$$

A set of this form is an *affine subspace*. It contains p but need not contain the zero column.

Theorem 18.1 (All solutions from one particular solution)**.** *Suppose $Ap = b$. Then the complete solution set of $Ax = b$ is*

$$\{x \in K^n : Ax = b\} = p + \ker A.$$

Proof. If $Ax = b$, linearity gives $A(x - p) = Ax - Ap = b - b = 0_m$. Thus $h = x - p$ belongs to $\ker A$ and $x = p + h$. Conversely, if $h \in \ker A$, linearity gives $A(p + h) = Ap + Ah = b + 0_m = b$. These two implications prove equality of the sets. ∎

Two solutions therefore differ by a homogeneous solution. Choosing another particular solution gives the same set, since the theorem identifies either translate with the solutions of the same system. If $b \neq 0_m$ and the system is consistent, its solution set is affine and is not a linear subspace: $A0_n = 0_m \neq b$ excludes the zero column. For $b = 0_m$, choose $p = 0_n$ and recover the null space itself.

For a consistent system, the solution is unique exactly when $\ker A = \{0_n\}$. If the null space contains a nonzero column h, the columns $p + th$, with $t \in K$, are distinct solutions. To check distinctness, equality for two parameters would give $(t_1 - t_2)h = 0_n$; a nonzero entry of h forces $t_1 = t_2$. Thus a nonzero homogeneous direction gives infinitely many solutions.

Figure 18.1 shows the translate in a real two-dimensional picture.

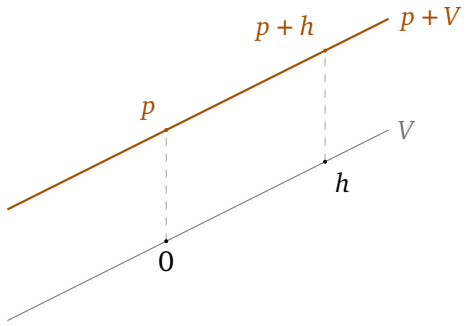


Figure 18.1: A real two-dimensional illustration of $V = \ker A$ and its translate. The corresponding points on the two parallel lines differ by the same column p.

18.3 Span, independence and coordinates

For columns $v_1, \dots, v_k \in K^n$, their *span* is the set of all linear combinations

$$\operatorname{span}(v_1, \dots, v_k) = \left\{ \sum_{j=1}^{k} c_j v_j : c_j \in K \right\}.$$

A span is a linear subspace. The zero choice of all coefficients gives 0_n. Adding two combinations adds their coefficients, and multiplying by a scalar multiplies each coefficient. Thus both closure properties hold. The span of the empty family is defined to be $\{0_n\}$, using the empty sum.

The family is *linearly independent over* K if

$$\sum_{j=1}^{k} c_j v_j = 0_n \quad \Longrightarrow \quad c_1 = \cdots = c_k = 0.$$

Otherwise it is *linearly dependent*. These extend the real definitions of Lecture 12 to the chosen column scalars. The empty family is independent, since there is no nonzero coefficient choice. An *ordered basis* of a subspace V is a finite ordered independent family whose span is V. We use the word basis with this ordering understood. The empty family is a basis of $\{0_n\}$.

Every column in a subspace with basis $v_1, \dots, v_k$ has a unique expression $x = \sum_{j=1}^{k} c_j v_j$. Existence follows from spanning. For uniqueness, subtract two expressions to get $\sum_j (c_j - d_j) v_j = 0_n$. Independence forces $c_j = d_j$ for each j. The coefficients in that order are the *coordinates in the basis*. With an empty basis, the zero column has the single empty representation. The standard coordinate columns form a basis of K^n by the same component argument as in Lecture 12.

To test membership in a span, form the coefficient matrix whose columns are $v_1, \dots, v_k$ and solve for the unknown coefficients c_j. To test independence, set the target column to zero. Independence means that this homogeneous coefficient system has only its zero solution. These are elimination tests from Lecture 17.

18.4 The basis supplied by free variables

In Lecture 17, the consistent family at $\lambda = \mu = 0$ was

$$x(t) = \begin{pmatrix} -9 \\ 6 \\ 0 \end{pmatrix} + t \begin{pmatrix} 7 \\ -3 \\ 1 \end{pmatrix}.$$

The difference between any two solutions is a multiple of $(7, -3, 1)^T$. Setting the right side of that system to zero gives precisely these multiples: the free coordinate is t, and choosing it as 1 produces the basis column. The construction now extends this idea to any number of free coordinates.

Reduce $[A \mid 0_m]$ to reduced echelon form. The augmented column remains zero under every row operation. Let the coefficient pivots in this chosen reduction occupy columns $\ell_1, \ldots, \ell_r$ and the free columns be $f_1, \ldots, f_k$, where $k = n - r$. Write r_{ij} for the entries of the reduced coefficient matrix. Its nonzero equations are

$$x_{\ell_i} = -\sum_{j=1}^{k} r_{if_j} x_{f_j}, \qquad i = 1, \ldots, r.$$

For each $j = 1, \ldots, k$, let h_j be the homogeneous solution whose free coordinate at f_j is 1 and all its other free coordinates are 0. The pivot equations determine the remaining entries. The existence and uniqueness for this assigned free list follow from Theorem 17.3.

> **Theorem 18.2** (Null-space basis from free variables). *The special columns $h_1, \ldots, h_k$ form a basis of* $\ker A$. *Every homogeneous solution has a unique representation $h = t_1 h_1 + \cdots + t_k h_k$, and t_j is its coordinate in the free position f_j. If there are no free columns, the empty family is a basis of* $\ker A = \{0_n\}$.

Proof. First suppose $k \geq 1$. A relation $\sum_j c_j h_j = 0_n$ has value c_ℓ in free position f_ℓ, by the way the special columns were assigned. Thus $c_\ell = 0$ for each ℓ, proving independence.

For a homogeneous solution h, choose $t_j = h_{f_j}$. The combination $\sum_j t_j h_j$ is homogeneous by linearity and has exactly the same free coordinates as h. A given free assignment has exactly one homogeneous solution by Theorem 17.3, so this combination equals h. This proves spanning. Independence gives uniqueness of its coefficients.

If $k = 0$, there are no free variables, and the homogeneous system has only 0_n by the same consistency theorem. The empty independent family spans this set, proving the final case. ∎

The construction proves a basis for each null space without requiring a basis-existence theorem for arbitrary subspaces. There are at most m coefficient pivots in any reduction. Consequently, if $n > m$, there is a free column and a nonzero homogeneous solution: assign 1 in that free coordinate and 0 in the others. For nonzero right sides, consistency is still a separate condition. The last task on Sheet 17 gives an explicit inconsistent example with more unknowns than equations.

Rank and the dimension convention

The cited pivot-count invariance in Theorem 17.4 makes the number of coefficient pivots independent of the chosen reduction. This number is the *rank* of A, denoted by $\operatorname{rank} A$. For

null spaces, the *dimension convention* used here is

$$\dim \ker A = n - \operatorname{rank} A. \tag{18.3}$$

The constructed free-variable basis has exactly that many columns. The count's independence from the reduction route uses the cited result; the basis construction and its independence and spanning proofs above hold directly for any chosen reduction. This convention specifies dimension for null spaces through elimination.

18.5 A complete affine family

Consider $Ax = b$ with

$$A = \begin{pmatrix} 1 & 2 & -1 & 3 \\ 2 & 4 & 1 & 5 \end{pmatrix}, \qquad b = \begin{pmatrix} 4 \\ 9 \end{pmatrix}.$$

Subtract $2R_1$ from R_2, divide the new second row by 3 and add it to the first row. These operations give the reduced augmented matrix

$$\left(\begin{array}{cccc|c} 1 & 2 & 0 & 8/3 & 13/3 \\ 0 & 0 & 1 & -1/3 & 1/3 \end{array}\right).$$

The pivot columns are 1 and 3; the free columns are 2 and 4. Set the free variables to zero to obtain the particular solution

$$p = \begin{pmatrix} 13/3 \\ 0 \\ 1/3 \\ 0 \end{pmatrix}.$$

Its two original left sides are $13/3 - 1/3 = 4$ and $26/3 + 1/3 = 9$. For the homogeneous system, assign $(x_2, x_4) = (1, 0)$ and then $(0, 1)$. The resulting basis columns are displayed together as

$$[\, h_1 \; h_2 \,] = \begin{pmatrix} -2 & -8/3 \\ 1 & 0 \\ 0 & 1/3 \\ 0 & 1 \end{pmatrix}.$$

For h_1, the original left sides are $-2+2 = 0$ and $-4+4 = 0$. For h_2, they are $-8/3-1/3+3 = 0$ and $-16/3 + 1/3 + 5 = 0$. Their free-coordinate pairs are $(1, 0)$ and $(0, 1)$, which also verify independence directly.

By Theorems 18.1 and 18.2, every solution is

$$x = p + sh_1 + th_2 = \begin{pmatrix} 13/3 - 2s - 8t/3 \\ s \\ 1/3 + t/3 \\ t \end{pmatrix}, \qquad s, t \in K.$$

The free coordinates recover s and t, so no two parameter pairs give the same solution. There are no further solutions because every difference from p belongs to the null space spanned by these columns. The rank is 2 and the null-space dimension convention gives $4 - 2 = 2$. The affine solution set has the same two independent parameters. It does not contain 0_4, since $b \neq 0_2$.

18.6 One span and its coordinates

Let v_1, v_2 and v_3 be the columns of

$$B = \begin{pmatrix} 1 & 2 & -1 \\ 2 & -1 & 8 \\ -1 & 3 & -9 \end{pmatrix}.$$

To describe their span, solve $Bc = x$ for a given target $x = (x_1, x_2, x_3)$, interpreted as a column. The coefficient unknowns are c_1, c_2 and c_3. Apply $R_2 \leftarrow R_2 - 2R_1$, then $R_3 \leftarrow R_3 + R_1$ and $R_3 \leftarrow R_3 + R_2$. Divide R_2 by -5 and clear its pivot above. The resulting row-equivalent array is

$$\left(\begin{array}{ccc|c} 1 & 0 & 3 & (x_1 + 2x_2)/5 \\ 0 & 1 & -2 & (2x_1 - x_2)/5 \\ 0 & 0 & 0 & x_2 + x_3 - x_1 \end{array}\right).$$

It is consistent exactly when $x_1 - x_2 - x_3 = 0$. Thus the span is the set of targets satisfying that scalar equation. For real columns, it is the plane through the origin with normal $(1, -1, -1)$, in agreement with Lecture 13.

Setting the target to zero gives coefficient solutions $c_1 = -3t$, $c_2 = 2t$, $c_3 = t$. The nonzero choice $t = -1$ yields the relation $3v_1 - 2v_2 - v_3 = 0$. The three-column list is therefore dependent. The first two columns are independent: when $c_3 = 0$ and the target is zero, the first two reduced rows force $c_1 = c_2 = 0$. They also span the full displayed target set, since one can take $c_3 = 0$ whenever the consistency condition holds. Hence v_1, v_2 form an ordered basis of the same span.

For the target

$$w = \begin{pmatrix} 0 \\ 5 \\ -5 \end{pmatrix},$$

the consistency expression is $0 - 5 - (-5) = 0$. Its basis coordinates are $(x_1 + 2x_2)/5 = 2$ and $(2x_1 - x_2)/5 = -1$. Thus $w = 2v_1 - v_2$. In the original dependent three-column list its representations are $c_1 = 2 - 3t$, $c_2 = -1 + 2t$, $c_3 = t$ with $t \in K$. Their nonuniqueness comes from the explicit relation among the three columns. The standard first coordinate column is not in this span, since $1 - 0 - 0 \neq 0$.

Summary 18.

- Equations (18.1) and (18.2): the homogeneous solution set is a linear subspace, by linearity of the coefficient action.
- Theorem 18.1: a consistent system has solution set $p + \ker A$ for any particular solution p. It is unique exactly when $\ker A = \{0_n\}$.
- An ordered basis is independent and spans its subspace. Its coordinate coefficients are unique. Span and independence tests are coefficient systems.
- Theorem 18.2: one special homogeneous solution for each free variable gives a basis. The empty family handles a zero null space.
- Equation (18.3): rank and the null-space dimension convention use the cited route-independent pivot count from Lecture 17. For an actual reduction with r pivots, the proved basis has $n - r$ columns.

Sheet 18 tests complete affine families, basis coordinates and the dimension convention. Lecture 19 develops matrix operations and their relation to composition.

References

D. C. Lay, S. R. Lay and J. J. McDonald, *Linear Algebra and Its Applications*, 5th ed., Pearson, 2016, §§1.5,1.7 and 4.3. S. Lang, *Introduction to Linear Algebra*, 2nd ed., Springer, 1986, Chapter III, §4.